SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE: CSA13**

COURSE OUTCOME COVERED

CO1: ***Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)***

Assessment Tool Weightage: 10%

QUESTIONS: 4 **Total Marks:20**
Duration : 1 Hour

Technical Presentation

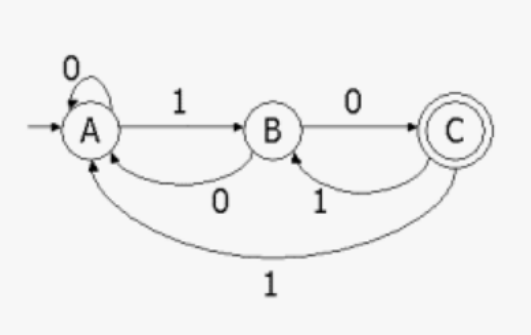
D.Naveen (192372177)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

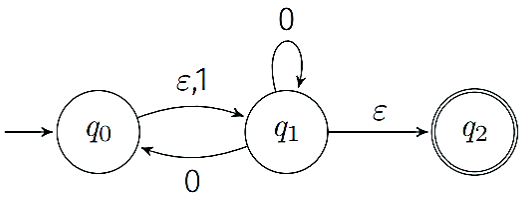
1.How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet Σ={0,1}. Construct a DFA that represents the same language L



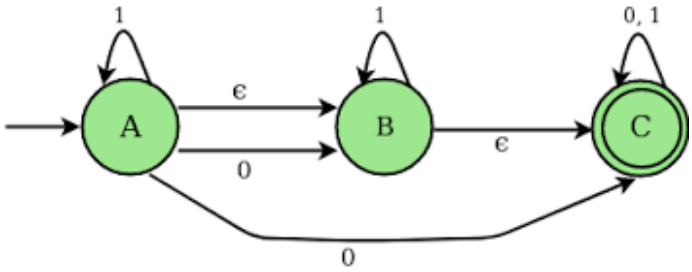
2.How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

3.What role do epsilon (ε) transitions play in finite automata, and how can ε-NFA be converted into an equivalent NFA without epsilon transitions? construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states



4.A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS – THEORY OF COMPUTATION (CSA13)

Name: D. Naveen **Reg. No:** 192372177

1. Conversion of NFA to equivalent DFA

An NFA is converted to a DFA by the **subset (powerset) construction**. Each DFA state represents a set of NFA states. Start with the set containing the NFA start state, compute the union of transitions for each input symbol, and repeat until no new sets appear. Any DFA set containing an NFA final state is final. Challenges include possible exponential growth (up to 2■ states), tracking state sets carefully, and handling missing transitions by using the empty set as a dead state.

For the given NFA, the transitions read from the diagram are: A --0→ A, A --1→ B, B --0→ A and C, C --1→ A and B. The start state is A and C is final.

DFA state (subset)	Input 0	Input 1	Final?
→ {A}	{A}	{B}	No
{B}	{A,C}	∅	No
* {A,C}	{A}	{A,B}	Yes
∅ (dead)	∅	∅	No
{A,B}	{A,C}	{B}	No

→ = initial state; * = accepting state. The accepting DFA state is any subset containing C.

2. Mathematical induction proof

We prove that for every positive integer n: $1^2 + 2^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$.

Base case (n = 1): Left side = $1^2 = 1$. Right side = $1(2)(3)/6 = 1$. Both sides are equal, so the statement is true for n = 1.

Inductive hypothesis: Assume the formula is true for n = k, i.e. $\Sigma(i^2)$, $i=1\dots k = k(k + 1)(2k + 1)/6$.

Inductive step: For n = k + 1,

$\Sigma(i^2)$, $i=1\dots k+1 = k(k + 1)(2k + 1)/6 + (k + 1)^2$
$= (k + 1)[k(2k + 1) + 6(k + 1)]/6$
$= (k + 1)(2k^2 + 7k + 6)/6$
$= (k + 1)(k + 2)(2k + 3)/6$
$= (k + 1)((k + 1) + 1)(2(k + 1) + 1)/6$.

Thus the formula holds for k + 1. By mathematical induction, it is true for all positive integers n.

3. Removing ε-transitions from the given ε-NFA

An ε-transition lets an automaton change state without consuming an input symbol. To remove ε-transitions, find each ε-closure, use $\delta'(q,a) = \varepsilon\text{-closure}(\text{move}(\varepsilon\text{-closure}(q), a))$, and make a state final whenever its ε-closure contains an original final state.

From the diagram: q■ has an ε-edge to q■ and a 1-edge to q■; q■ has a 0-edge to q■ and a 0-loop; q■ has an ε-edge to q■. q■ is the original final state.

State	ε-closure	New transition on 0	New transition on 1	Final?
→ q■	{q■,q■,q■}	{q■,q■,q■}	{q■,q■}	Yes
* q■	{q■,q■}	{q■,q■,q■}	∅	Yes
* q■	{q■}	∅	∅	Yes

Initial state: q■. New final states: q■, q■ and q■, because each ε-closure contains q■. ∅ means no reachable state.

4. ε-closures and equivalent DFA

The ε-edges are A → B and B → C. Therefore:

State	ε-closure
A	{A,B,C}
B	{B,C}
C	{C}

Using subset construction with ε-closures, the reachable DFA states are {A,B,C}, {B,C}, and {C}. The start state is {A,B,C}; any subset containing C is accepting.

DFA state	Input 0	Input 1	Final?
→ * {A,B,C}	{B,C}	{A,B,C}	Yes
* {B,C}	{C}	{B,C}	Yes
* {C}	{C}	{C}	Yes

Transition interpretation: A has a 1-loop and transitions on 0 to B and C; B has a 1-loop; C loops on both 0 and 1. Initial state is {A,B,C}. All listed DFA states are final.

Note: The answers above are based on the transitions visible in the attached diagrams. Please compare the transition labels with your original copy before submission.